

Daily Briefing App for Indian News: Live RSS, Zero-shot Classification, and LLM Summarization

Your Name

Abstract

This report presents a daily-briefing system that aggregates Indian news from live RSS feeds, classifies stories into five categories using a zero-shot model, and generates concise three-line summaries with an LLM. We evaluate classification on a long-horizon Indian headlines corpus and analyze runtime behavior under caching. We also include an ablation study on prompt-template sensitivity and input variants.

1 Introduction

State the motivation, problem definition, and contributions.

2 Data

2.1 Live RSS Sources

The Hindu, NDTV, and Indian Express RSS feeds are polled at app load time.

2.2 Static Evaluation Corpus

Use the Kaggle India headlines dataset (21 years) and map labels to the target taxonomy.

3 Method

3.1 Pipeline

RSS fetch → text cleaning → zero-shot classification → LLM summarization → tabbed UI and newsletter generation.

3.2 Zero-shot Classifier

Model: `facebook/bart-large-mnli`. Candidate labels: Politics, Sports, Technology, Business, Entertainment.

3.3 Summarization

Model/API: Gemini 1.5 Flash. Prompt constrained to exactly three short lines.

3.4 Caching

Use Streamlit cache for model loading, RSS responses, and summary generation.

4 Results

4.1 Classification Metrics

Provide accuracy, macro-F1, per-class metrics, and confusion matrix.

4.2 Qualitative Summaries

Add examples of article text and generated summaries.

4.3 Latency

Report end-to-end latency with and without cache.

5 Ablation Study

Include at least two ablations:

- Hypothesis template variants in zero-shot classification.
- Title-only vs title+description inputs.
- Cache on/off latency comparison.

6 Limitations

Discuss RSS quality, label ambiguity, API quotas, and summary factuality risks.

7 Conclusion

Summarize findings and future work.

References

1. BART-MNLI model card.
2. Gemini API documentation.
3. Streamlit documentation.
4. Feedparser documentation.
5. Kaggle India headlines dataset citation.